Synthetic Cultural Agents 2.0: Beyond Persona-Prompting

A Position Paper on Structural Preference Estimation, Measurement, and the Validity of Cultural Constructs in Language Models

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Status. Intermediate internal summary of SCA 2.0: structural conjecture, implemented DPO pipeline, Tier-1 synthetic pair recovery, Tier-2 out-of-sample evaluation on World Values Survey Wave 7 items ($N = 35$), and a construct-validity layer that identifies admissible measures of GPS preference dimensions by partial identification. Not a submission draft. Code & data: github.com/EconLLM-Lab/SCA2_PofW; measurement instrument: github.com/Bonorinoa/cvprofiles.

Third-party evaluation required before any submission positioning is adopted. Current empirical results are **proof-of-concept**: the adapter layer covers two adapters and two countries, and expanding the country panel and the menu of AI-generated measures is required before any headline claim. Numbers are preliminary and included for methodological exposition.

Statement of Significance

Economics has always studied latent constructs—preferences, trust, patience, social preferences—but its instruments for observing them are costly, sparse, and each is an imperfect representation of the underlying attribute. Which survey question you use to measure "trust" can change the answer you get about how trust relates to development as much as the phenomenon itself does. As AI makes candidate measures cheap, the binding constraint is no longer building measures; it is knowing which ones are scientifically admissible. This paper develops two open-source instruments: (i) a pipeline that instantiates a measured cultural state in a language-model policy, producing low-cost synthetic populations for experimental and comparative-statical analysis; and (ii) a partial-identification tool (`cvprofiles`) that, given a researcher's nomological network, selects the set of admissible measures of a latent construct and reports the range of any downstream estimate consistent with all admissible measures. Applied to the Global Preferences Survey and the World Values Survey, the framework shows that only a minority of candidate survey measures are admissible representations of the six GPS preference dimensions, and that the canonical single-item trust measure and a multi-item composite agree on average trust but disagree on *who* trusts—with direct consequences for hypothesis generation, pre-field survey design, and cross-country policy analysis.

Abstract

Large language models are increasingly used as simulators of human judgment in economics and adjacent social sciences, yet their default alignment reflects predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) preference data. Earlier Synthetic Cultural Agents (SCA 1.0) addressed this gap by conditioning frontier models on ethnographic profiles at inference time—useful for piloting but reduced-form: cultural signal lives only in the prompt, is fragile to model updates, and supports no formal identification.

This position paper outlines SCA 2.0, a structural alternative built on two fused contributions. *First*, an open-source estimation pipeline: Direct Preference Optimization (DPO) is treated as maximum-likelihood estimation in a Bradley–Terry random-utility model; the Global Preferences Survey (GPS) six-dimensional cultural state vector generates culture-conditioned synthetic preference data; a student model (`Llama-3.1-8B-Instruct`) is fine-tuned via QLoRA with unconditioned prompts so that cultural dispositions are encoded into parameters rather than transient context; and the resulting agents are evaluated out-of-sample on human survey data—an open pipeline for synthetic experiments on cultural states. *Second*, a measurement instrument: we formalize the Cronbach–Meehl nomological network as a partial-identification problem over a menu of candidate measures, implemented in the open-source package `cvprofiles` (score, restrict, identify, report). Given a researcher-supplied network of moment restrictions with literature-anchored thresholds, the tool computes slack profiles, the admissible set M^*, and the construct-identified range $[L,U]$ of any downstream target. We demonstrate the machinery on the trust dimension of the GPS/WVS bridge: only three of four WVS trust facets are admissible representations of GPS trust; the canonical single item (Q57) is rejected at the stated threshold; and a "survivor composite" reproduces GPS trust's demographic structure while the single item does not—evidence that representation choice is consequential, not cosmetic. Empirically, the MEX adapter achieves strong distributional alignment with Mexico's WVS responses (matched-better on 97.1% of 35 items, $\Delta\text{TVD} = 0.1495$, 95% CI $[0.0964, 0.2152]$), while the USA adapter degrades fit relative to the base model with mechanism unresolved. We delineate reliable applications (relative comparative statics, pre-field instrument stress-testing, measurement-fragility quantification) from unreliable ones (absolute headcount polling, fine-tuning over already-dominant priors), and discuss frontier applications—power analysis without priors, survey-item pre-piloting, construct induction for arbitrary models, and country- or individual-level response simulation.

Keywords: cultural preferences; Direct Preference Optimization; Bradley–Terry; Global Preferences Survey; structural estimation; construct validity; partial identification; World Values Survey

1. Introduction

Social scientists increasingly use large language models as low-cost simulators of survey responses, experimental subjects, and policy interlocutors [19, 2, 1]. The practical appeal is clear: a model can generate thousands of responses in minutes, enabling exploratory design before expensive fieldwork. The scientific risk is equally clear. Preference fine-tuning and human feedback data for frontier models are concentrated in WEIRD populations [18, 16]. When a model is asked to "answer as someone in Mexico City," its behavior still largely reflects that default alignment—unless culture is introduced in a disciplined way.

The first generation of Synthetic Cultural Agents [10] showed that persona prompting can reproduce non-WEIRD patterns in economic experiments to a surprising degree. For piloting and hypothesis generation, that result is valuable. Methodologically, however, persona prompting is a reduced-form intervention: cultural information lives in the context window, vanishes when the session ends, and provides no structural object (preferences, technology, information) that can be identified, tested, or counterfactually varied.

SCA 2.0 reframes the problem as *structural preference estimation* under *measurement discipline*. The core conjecture is:

> *Cultural preferences measured by the Global Preferences Survey can be encoded into language-model policies via preference fine-tuning, so that implied utilities recover the direction of culture-conditioned choices and support formal evaluation against independent survey moments.*

The paper makes two contributions, which are deliberately coupled. The **first contribution** is an open-source pipeline for what we call *synthetic experiments on cultural states*: GPS-grounded synthetic preference pairs, DPO fine-tuning of an open-weight student model with unconditioned prompts, and out-of-sample evaluation on human survey responses. Because the cultural state enters only through the training labels, the pipeline allows an analyst to hold the prompt design fixed while varying the state—a controlled intervention on the training distribution, not a randomized treatment of human culture. The **second contribution** is a measurement instrument: because the scientific status of the whole enterprise hinges on whether our measures (the survey facets, the composites, and the adapters themselves) actually measure the constructs they claim to, we formalize the classical construct-validity framework of Cronbach and Meehl [12] as a *partial-identification* problem [22] over a finite menu of candidate measures. The instrument is implemented in the open-source package `cvprofiles` (`SCORE` $\rightarrow$ `RESTRICT` $\rightarrow$ `IDENTIFY` $\rightarrow$ `REPORT`): given a researcher-authored nomological network of moment restrictions with thresholds anchored in the published literature, it computes slack profiles, the admissible set M^*, and the construct-identified range $[L,U]$ of any downstream target functional.

The methodological spine of the paper is a set of *equivalences across literatures* that together formalize AI as an economic measurement instrument. First, DPO training is Bradley–Terry maximum likelihood [27, 7]: fine-tuning a policy on preference data is structural estimation of a random-utility model, so the trained policy's implied reward margins are utility differences. Second, option-likelihood scoring is Luce–Bradley–Terry–McFadden choice probability [24]: the model's softmax over response options is a choice-probability simulator—a measurement function $m_{AI}(X)$ over the space of prompts and responses. Third, the Cronbach–Meehl nomological network is a system of moment inequalities [12, 22, 14]: validating a measure is computing an identified set. None of these equivalences is new within its own literature; their *fusion* is. The result is a general system of latent structural modeling with two components—a preference-alignment component (the pipeline) and a validation component (the measurement instrument)—of which cultural modeling is the first application. The SCA research agenda's fundamental objective is the *measurement of culture*; this paper argues that the route runs through the general instrument, instantiated on culture, rather than through culture-specific engineering.

The connection between the two literatures is the methodological spine of the paper. Construct validity (psychometrics) answers *which measures are admissible representations of a latent construct*; partial identification (econometrics) answers *what set of conclusions survives all admissible representations*. We show that the Cronbach–Meehl nomological network is naturally formalized as a system of moment inequalities—the same formal object at the center of the partial-identification literature—and that this formalization, implemented in `cvprofiles`, converts "which survey item should I use?" from a judgment call into an identified set.

The empirical vehicle is the GPS–WVS bridge. The GPS [16] supplies incentivized, behaviorally validated measures of patience, risk taking, positive and negative reciprocity, altruism, and trust across 76 countries. The World Values Survey supplies candidate survey measures of the same constructs. We evaluate (i) whether the WVS facets are admissible representations of the GPS dimensions, at country level and at demographic-structure level; (ii) whether DPO adapters trained on GPS-grounded synthetic data reproduce the empirical response distributions of their target countries on unseen WVS items; and (iii) whether the choice of representation changes downstream conclusions—it does.

The paper proceeds as follows. Section 2 reviews the three literatures the framework spans. Section 3 presents the methods: the estimation pipeline and the measurement framework, with the formal specification of the network and a worked applied example on trust. Section 4 reports the empirical results across both tiers and the construct-validity layer. Section 5 discusses reliable and unreliable applications, frontier applications, and the broader positioning of rigorous measurement in developing economics. Section 6 outlines next steps.

2. Related Work

2.1 LLMs as simulated subjects and synthetic populations

A growing literature uses LLMs as simulated respondents [19, 2, 1]; much of it is reduced-form role prompting, the family to which SCA 1.0 belongs [10]. Complementary work documents cultural gaps and representation asymmetries in model outputs [9, 15] but typically stops at evaluation rather than estimation. Recent econometric treatments of LLM measurement [21, 11] frame model outputs as variables subject to measurement error and missing-data problems, reviving the validation-sample machinery of [6]. Our pipeline contributes the estimation side: rather than treating the model as an instrument to be corrected, we *train* the cultural state into the policy, then validate the policy against human data.

2.2 Preference estimation with LLMs

DPO [27] optimizes a policy so preferred responses receive higher probability than dispreferred ones relative to a reference model; under standard assumptions this coincides with Bradley–Terry maximum likelihood for a scaled log-probability score [7, 24]. We use this equivalence as a license to speak econometrically: the trained policy's implied reward margins are utility differences under the country-specific cultural state, identified up to scale and location. This structural reading of DPO is the bridge between the machine-learning pipeline and the measurement framework.

2.3 Construct validity and the nomological network (psychometrics)

Validity was classically defined by correlation with a criterion or by content judgment; both fail when the attribute is latent and no criterion exists. Cronbach and Meehl [12] resolved the crisis by introducing the *nomological network*: a system of theoretical laws linking a construct to other constructs and to observables, so that validity becomes a matter of testing the network's implied relations. Campbell and Fiske [8] supplied the operational template—convergent validity (measures of the same trait, ideally by different methods, cohere) and discriminant validity (measures of different traits do not collapse). Messick [25] unified the aspects of validity around score interpretation; Borsboom, Mellenbergh and van Heerden [5] added the crucial realism caveat: a network establishes *consistency*, not *causation*—the construct's existence and the direction of causation between attribute and score are not established by passing the network's restrictions.

2.4 Partial identification and measurement in econometrics

Leamer's extreme-bounds analysis [20] asked how much conclusions move across specifications; Sala-I-Martin's "two million regressions" [28] made the question unavoidable once computation became cheap. Manski [22, 23] formalized the discipline: report the *set* of parameter values consistent with the data and the stated assumptions—no more, no less. The econometric measurement literature supplies the treatment of mismeasured variables and multi-measure systems [6, 29], and the recent LLM-as-measurement framework of Ludwig, Mullainathan and Rambachan [21] connects these tools to frontier models. Dell and Rambachan's 2026 NBER Methods Lecture [14] argues that AI makes candidate measures cheap and shifts the bottleneck to *choosing among many plausible representations*, which may yield different estimates of the target parameter; they propose formalizing the Cronbach–Meehl network as a partial-identification problem—the approach implemented here.

2.5 The synthesis: construct validity as partial identification

The two literatures answer different questions with different machinery, but they are the same question at different levels. Construct validity asks: *which measures are admissible representations of the construct, given theory?* Partial identification asks: *what can be learned about the target parameter, given data and assumptions?* A nomological network is a set of assumptions; its restrictions are moment inequalities; the admissible set M^* is the identified set for the "measure" object; and the image of a downstream target β

on M^* is the identified set for the conclusion. Our contribution at the methodological level is to make this equivalence operational in a reproducible open-source instrument (`cvprofiles`) and to demonstrate it on a real latent construct—country-level economic preference disposition—where the menu of candidate measures comes from two independent survey traditions.

3. Methods

3.1 The synthetic-agent pipeline: synthetic experiments on cultural states

3.1.1 Primitives.

Let x denote a decision scenario (prompt) and y a free-text response. A cultural state is a vector

$$z = \big(\delta,\ \gamma,\ \xi^+,\ \xi^-,\ \alpha,\ \tau\big)^\top \in \mathbb{R}^6, \tag{1}$$

corresponding to time preference (patience), risk taking, positive reciprocity, negative reciprocity, altruism, and trust from the GPS. An SCA is a policy $\pi(\cdot \mid x; z)$ whose behavior is governed by a separable reward

$$R(x,y;z) \;=\; r_{\text{task}}(x,y) \;+\; \phi(z)^\top m(x,y), \tag{2}$$

where r_{task} captures generic competence (coherence, relevance), $m(x,y) \in \mathbb{R}^6$ is a vector of measurable behavioral features of the response, and $\phi : \mathbb{R}^6 \to \mathbb{R}^6$ maps GPS scores into reward weights.

The structural assumption. Equation (2) is a *structural working assumption* (not an estimated fact): it claims that the country-specific component of an agent's behavior is a linear function of six externally measured preference traits, mediated by response-scoring features. Two testable implications follow. First, *separability*: the task term r_{task} is country-invariant, so any cross-country behavioral difference must load on $\phi(z)^\top m(x,y)$. Second, *dimensional structure*: only six dimensions of cultural variation are permitted to matter. Both implications are falsifiable—the first by the matched-versus-cross evaluation design (Section 4), the second by out-of-sample items that "should not" be explained by any dimension.

3.1.2 Choice model.

We treat the agent's response as a discrete choice over a finite option set $\mathscr{Y} = \{y_1, \ldots, y_K\}$ (the response codes of a survey item or the two completions of a preference pair). A Luce–Bradley–Terry–McFadden random-utility model sets

$$\Pr(y_k \mid x; z) \;=\; \frac{\exp\big(R(x,y_k;z)\big)}{\sum_{j=1}^{K} \exp\big(R(x,y_j;z)\big)}, \tag{3}$$

which is the softmax over option rewards. For a pair $\{y^w, y^\ell\}$, equation (3) implies the logistic paired-comparison probability

$$\Pr(y^w \succ y^\ell \mid x) \;=\; \sigma\big(R(x,y^w;z) - R(x,y^\ell;z)\big), \tag{4}$$

with $\sigma(u) = (1+e^{-u})^{-1}$. Equation (3) is the idealized object approximated by the evaluation pipeline: the option-likelihood protocol (Section 4) computes $\hat{p}_k \propto \exp(\text{logprob}(y_k \mid x))$ over all valid response codes and compares the resulting distribution to the empirical population distribution. The LLM is therefore evaluated as a *choice-probability simulator*, not treated as a calibrated text generator.

3.1.3 DPO as structural estimation.

Given preference data $\{(x_i, y_i^w, y_i^\ell)\}_{i=1}^n$ in which y^w is preferred to y^ℓ under cultural state z, the Bradley–Terry likelihood is

$$\mathscr{L}(R) \;=\; \prod_{i=1}^{n} \sigma\big(R(x_i, y_i^w; z) - R(x_i, y_i^\ell; z)\big). \tag{5}$$

Under the Bradley–Terry and KL-regularized-RL assumptions used to derive DPO, Direct Preference Optimization [27] has this likelihood form while reparameterizing reward differences through the log-ratio of the trained policy to a frozen reference policy:

$$R(x, y_1; z) - R(x, y_2; z) \;=\; \beta\left[\log\frac{\pi_\theta(y_1 \mid x)}{\pi_{\text{ref}}(y_1 \mid x)} - \log\frac{\pi_\theta(y_2 \mid x)}{\pi_{\text{ref}}(y_2 \mid x)}\right], \tag{6}$$

where π_{ref} is a frozen reference model and $\beta > 0$ is a scale temperature. Equation (6) supplies an empirical reward-margin proxy for the structural reward in equation (2): after training, the analyst can compute, for any held-out pair, the *implied reward margin* $\widehat{\Delta r} = \beta\,[\text{log-ratio of chosen} - \text{log-ratio of rejected}]$. We interpret this as a model-implied utility difference only under the maintained assumptions; it is not a separately identified human utility.

Identification and its limits. Equation (6) identifies reward *differences* up to scale (β) and location. The two structural features we *can* recover are (i) the sign of Δr and (ii) the own-versus-other contrast: whether an adapter trained on country c's labels assigns systematically larger margins to c's chosen options than an adapter trained on c'. We cannot recover the absolute level of R nor, in general, the individual coordinates of $\phi(z)$ without additional moment conditions.

3.1.4 Pipeline stages.

Figure 1 summarizes the pipeline: (A) the cultural state z_c is read from the GPS country vector; (B) country-neutral scenarios are generated across five facets per dimension; (C) preference pairs are constructed by labeling fixed high/low option sets with the country's state; (D) quality control scores each pair on the six dimensions (monotonicity, distance, contamination); (E) a student model is fine-tuned via QLoRA with unconditioned prompts. Validation proceeds through Tier-1 (implied Δr on held-out synthetic pairs), Tier-2 (out-of-sample WVS option likelihood), and the construct-validity layer of Section 3.2.

3.2 The measurement framework: construct validity as partial identification

3.2.1 The problem.

The pipeline above produces agents whose behavior we interpret as a model-implied representation of country c's preference state. Every link in that interpretation passes through *measurement*: the GPS state is a measure; the WVS items are measures; the adapter's option distributions are measures. None of them is the latent attribute itself. The scientific question—*which of these candidate measures are admissible representations of the construct?*—is precisely the question the construct-validity tradition addresses, and we answer it with partial-identification machinery.

3.2.2 Formal specification.

Let X denote the underlying reality (respondents, countries, their behaviors) and let a *measurement function* $m_j(X)$ be any disciplined projection of X into a score. The researcher supplies:

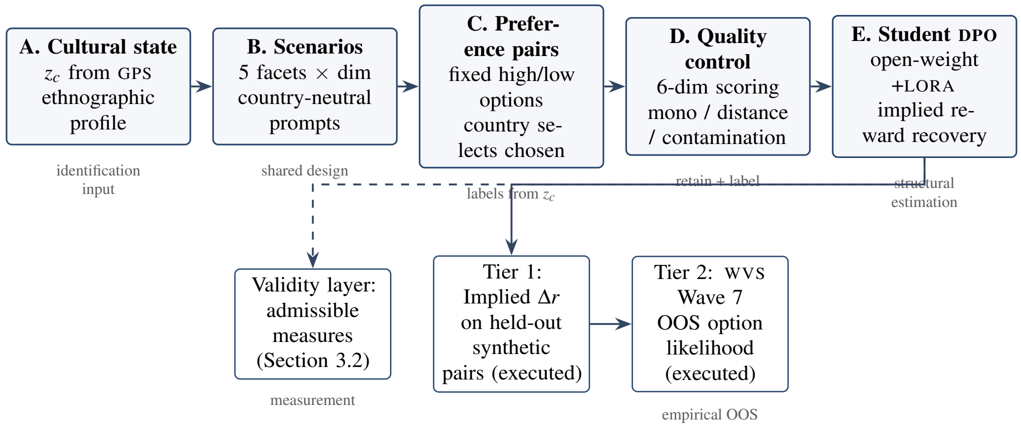

Figure 1: SCA 2.0 pipeline architecture. Culture enters as an external GPS vector z_c; synthetic pairs are generated, quality-controlled, and fine-tuned via DPO; validation proceeds through Tier-1 synthetic pair recovery, Tier-2 WVS option-likelihood evaluation, and the construct-validity layer of Section 3.2.

1. a **menu** $\mathscr{M} = \{m_j(\cdot)\}_{j\in\mathscr{J}}$ of candidate measures, already normalized so that both measures and downstream estimands are comparable across candidates;
2. a **nomological network**—a set $\mathscr{R}$ of moment restrictions, each stating a required behavior of any valid measure:
$$r \in \mathscr{R}: \qquad \mathbb{E}\big[g_r(m(X), V; \theta_r)\big] \geq 0, \tag{7}$$
where $V = (Z, W, G, Q)$ collects the auxiliary observables implicated by the network: Z = related quantities (convergent evidence), W = distinct quantities (discriminant evidence), G = groups that should differ on the construct (known-groups evidence), Q = conditioning covariates; and θ_r is the restriction's threshold, *anchored in published evidence* rather than data-mined;
3. a **tolerance** $\delta \geq 0$ (default 0): a measure is admitted when all slacks satisfy $s_r(m) \geq -\delta$.

The *slack* of measure m on restriction r is the sample analogue of the left-hand side of equation (7):
$$s_r(m; \theta_r) \;=\; \widehat{\mathbb{E}}\big[g_r(m(X), V; \theta_r)\big], \tag{8}$$
with positive values indicating satisfied restrictions and negative values the magnitude of violation. The *admissible set* is
$$\widehat{\mathscr{M}}^{*}_{\mathscr{R},\theta} \;=\; \{m \in \mathscr{M} : s_r(m; \theta_r) \geq -\delta \;\; \forall r \in \mathscr{R}\}. \tag{9}$$
This sample analogue encodes *conceptual* uncertainty conditional on the observed data and researcher-specified restrictions. It is a deterministic screening rule unless sampling uncertainty is propagated, as described below. A population identified set would replace the sample slacks with their population counterparts; neither object is exhaustive unless the candidate menu and restrictions are justified as sufficiently rich.

For a downstream target functional $\beta(\cdot)$ (a regression coefficient, a correlation, a treatment effect defined using the measure), the *construct-identified set* is
$$\widehat{\mathscr{B}}^{*}_{\mathscr{R},\theta} \;=\; \Big\{\beta(m) : m \in \widehat{\mathscr{M}}^{*}_{\mathscr{R},\theta}\Big\}, \tag{10}$$
and, for scalar targets, the headline report is the *construct-identified range*
$$[\widehat{L}, \widehat{U}] \;=\; \left[\inf_{m\in\widehat{\mathscr{M}}^{*}} \beta(m), \;\; \sup_{m\in\widehat{\mathscr{M}}^{*}} \beta(m)\right]. \tag{11}$$

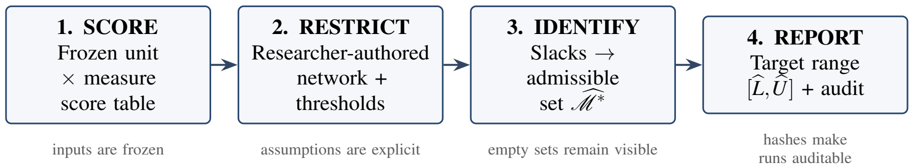

Figure 2: The measurement layer separates researcher judgment from computation. Researchers specify the candidate measures and nomological network; `cvprofiles` scores restrictions, computes the admissible set, propagates measurement uncertainty to downstream targets, and preserves the audit trail.

Inference and reproducibility. Sampling uncertainty is handled by bootstrap *over units* (the menu is fixed, not resampled), reporting percentile bands on the range endpoints over non-empty replicates; degenerate replicates are counted and reported. Threshold sensitivity is reported as a multiplicative λ-grid diagnostic (the headline is $\lambda = 1$; grid settings are excluded from the run identity). Paper-relevant numbers require frozen scores, pinned network/β, fixed seed, and package version, with a deterministic run id derived from their hash. Empty admissible sets and wide ranges are valid scientific outputs; thresholds are never loosened to avoid them.

Implementation. The framework is implemented in the open-source package `cvprofiles` [13] (`github.com/Bonorinoa/` whose state machine is exactly `SCORE` $\to$ `RESTRICT` $\to$ `IDENTIFY` $\to$ `REPORT`: it ingests a frozen unit×measure score table, binds the researcher-authored network and target functional, computes slacks and the admissible set, and emits JSON/HTML audit artifacts with run-id and hash discipline. The engine is score-agnostic and model-free; it contains no LLM calls. The network is always researcher-authored; the tool only evaluates.

3.3 Applied example: identifying admissible representations of GPS trust

We illustrate the machinery on the trust dimension of the GPS–WVS bridge. The construct is *country-level generalized trust*; the unit is a country, not an individual respondent. The menu is the four WVS Wave 7 trust facets—generalized (Q57), in-group (Q58, Q60), out-group (Q59, Q61–63), and institutional (Q64, Q69–71)—each scored from respondent-level items per a frozen protocol and aggregated to country means. This illustration uses the 42 countries in the WVS∩GPS overlap, distinct from the 35-question Tier-2 adapter evaluation sample. The nomological network is a single convergent restriction:

$$r_{\text{trust}}: \quad \text{Corr}(m, \text{gps\_trust}) \geq 0.30, \tag{12}$$

with threshold anchored in published survey–behavioral trust correlations ($r \approx 0.29$ in OECD, 2017 [26]; country-level survey–GPS correlations of 0.28–0.35 in prior work [16]). Table 1 reports the run (frozen protocol hash, pinned seed; run id `483fc619641476a9`).

Table 1: Construct-validity profile, trust dimension. Unit = country ($n = 42$). Restriction: $\text{Corr}(m, \text{gps\_trust}) \geq 0.30$ ($\delta = 0$). Slack $= r - 0.30$.

Measure	Items	Corr(m, GPS)	Slack	Verdict
trust-general (Q57)	Q57	0.278	-0.022	**rejected**
trust-in-group	Q58, Q60	0.349	$+0.049$	admissible
trust-out-group	Q59, Q61–63	0.303	$+0.003$	admissible (knife-edge)
trust-institution	Q64, Q69–71	0.332	$+0.032$	admissible
Survivor composite	in+out+inst	0.410	—	(derived reference)

The profile illustrates every element of the framework. The *slack column* shows that the single most-used trust item in the empirical literature, Q57 ("most people can be trusted"), is *rejected* at the literature-anchored threshold ($r = 0.278 < 0.30$), while three multi-item facets survive. The admissible set is $M^* = \{$in-group, out-group, institution$\}$; the survivor composite—their mean—correlates 0.41 with GPS trust. The θ-grid (Table 2) shows the identification is not knife-edge overall: at $\lambda = 0.5$ all four facets pass, at $\lambda = 1.0$ three pass, and at $\lambda = 1.5$ the set is *empty*—an honest report that the network plus data admit no measure at that standard.

Table 2: λ-grid diagnostic for the trust profile (multiplicative; headline $\lambda = 1.0$).

λ	threshold	$\|M^*\|$	admissible set
0.5	0.15	4	all four facets
1.0	**0.30**	**3**	in-group, out-group, institution
1.5	0.45	0	$\emptyset$ (empty M^* is an honest result)
2.0	0.60	0	$\emptyset$

The same machinery applied to the other five dimensions yields the outcome typology in Table 3: patience has a singleton admissible menu (only financial satisfaction survives), negrecip admits justifiability only, and risktaking, altruism, and posrecip have *empty* admissible sets at the headline threshold—the mapped WVS items are not admissible representations of those GPS constructs under this network and threshold. Across the full menu of 16 candidate facets, only 5 are admissible. This is the substantive content of the framework: *most survey measures of economic preferences are not admissible representations of the incentivized constructs*, and the framework says so in an auditable, threshold-transparent way.

Table 3: Outcome typology across the six GPS dimensions (P1 profiles, $\theta = 0.30$, $\delta = 0$).

Dimension	**Menu**	$\|M^*\|$	**Outcome**	**Admissible measures**
trust	4	3	nonempty	in-group, out-group, institution
patience	3	1	**point-identified**	financial (Q50)
negrecip	3	1	nonempty	justifiability (Q177, Q179)
risktaking	2	0	**empty**	—
altruism	1	0	**empty**	—
posrecip	3	0	**empty**	—
total	16	5	—	—

Finally, the framework's downstream object—the construct-identified range—is demonstrated by the `cvprofiles` H5 trust evaluation on public data ($n = 35$ countries; WVS$\times$GPS$\times$World Bank). The network there is richer: $\mathrm{corr}(m, \mathrm{gps\_trust}) \geq 0.3$, $\mathrm{corr}(m, \mathrm{rule\_of\_law}) \geq 0.3$, and $\mathrm{corr}(m, \mathrm{gini}) \leq -0.1$, with target $\beta(m) = \mathrm{Corr}(m, \log \text{ GDP per capita})$ (GDP deliberately excluded from the network). The result: $M^* = \{$trust-general, trust-in-group$\}$, both designed-invalid measures ($m_{\mathrm{noise}}, m_{\mathrm{share\_agriculture}}$) are rejected (false-admission rate $= 0$), and the construct-identified range of the trust–development correlation is

$$[L, U] = [0.371,\ 0.624], \tag{13}$$

with a units bootstrap band of $[0.174, 0.752]$ (17.5% empty replicates reported). The economic content: *how much of the trust–development relationship is sensitive to measurement choice is itself quantified*—a range of 0.25 correlation points across admissible measures, not a single number.

4. Empirical Results

4.1 Tier-1: Synthetic Preference Recovery

Table 4 reports Tier-1 results on held-out synthetic preference pairs ($n = 200$ per adapter–country cell). Both the Mexico and US adapters recover their own culture-conditioned synthetic labels well above chance (78.5% and 70.0% preference accuracy, respectively), while cross-country accuracy remains near chance (52.0% and

45.5%).[1]

Table 4: Tier-1 Cross-evaluation of country adapters on held-out synthetic preference pairs ($n = 200$ per cell). Preference accuracy is the share of pairs with positive implied reward margin ($\Delta r > 0$).

Adapter	Eval Data	n	Mean Δr	Pref. Accuracy	Mean P(chosen)
Mexico Adapter	Mexico	200	+1.70	0.785	0.714
Mexico Adapter	United States	200	−0.22	0.520	0.501
USA Adapter	Mexico	200	−0.20	0.455	0.464
USA Adapter	United States	200	+0.78	0.700	0.632

4.2 Tier-2: World Values Survey Out-of-Sample Evaluation

Tier-2 tests frozen adapters on $N = 35$ unseen Wave 7 WVS items (30 mapped to the six GPS dimensions; 5 demographic items) under unconditioned prompts (`PROMPT_COUNTRY_CONDITIONING = False`). Generated option distributions $\hat{P}$ are evaluated against survey-weighted human response distributions P_{WVS}. Table 5 summarizes the primary evaluation metrics.

Table 5: Tier-2 Out-of-Sample Evaluation on $N = 35$ unseen WVS Wave 7 survey questions under unconditioned prompts. Matched vs Cross (% Matched Better) reports the fraction of questions where the matched adapter achieves lower error than the cross adapter.

Target Population	Model / Adapter	TVD ↓	JSD ↓	ECE ↓	Matched vs Cross (% Matched Better)
Mexico Data (MEX WVS)	Base Model (`Llama-3.1-8B`)	0.5211	0.2165	0.0919	—
Mexico Data (MEX WVS)	**MEX Adapter (Matched)**	**0.4882**	**0.2010**	0.0967	**97.1**% (34/35)
Mexico Data (MEX WVS)	USA Adapter (Cross)	0.6377	0.2965	0.1220	—
USA Data (USA WVS)	**Base Model** (`Llama-3.1-8B`)	**0.3958**	**0.1439**	**0.0654**	—
USA Data (USA WVS)	MEX Adapter (Cross)	0.3888	0.1405	0.0719	—
USA Data (USA WVS)	USA Adapter (Matched)	0.5258	0.2246	0.0958	11.4% (4/35)

Key Finding 1: Distributional Alignment on Non-WEIRD Targets. On Mexico WVS data, the matched MEX adapter achieves strong distributional alignment, outperforming the cross USA adapter on 97.1% of questions (34/35 items). The mean difference in Total Variation Distance ($\mathrm{TVD}_{\mathrm{cross}} - \mathrm{TVD}_{\mathrm{matched}}$) is $+0.1495$ (95% CI: $[0.0964, 0.2152]$). Because prompts omit country names, this difference can only arise from the adapter weights: each adapter emits a single fixed response distribution, and the MEX adapter's fixed distribution sits closer to Mexico's observed WVS responses than the USA adapter's. This is evidence that country-specific dispositional signal was internalized into the weights during fine-tuning; it does not demonstrate that an adapter conditions on or "knows" its target country at inference time.

Plain-language reading. If the adapter weights were doing nothing, the matched and cross adapters would perform equally well and the split would be roughly 50/50; a win rate of 34/35 is far from that benchmark, with a confidence interval that does not cross zero. The result is directional, not calibrated: we do not claim that the adapter predicts the percentage of Mexicans who trust their family, only that it ranks trust patterns in the direction the empirical distribution supports.

Key Finding 2: USA Adapter Degradation (mechanism unresolved). On USA WVS data, the matched USA adapter performs *worse* than both the base model (TVD = 0.5258 vs. 0.3958) and the cross MEX adapter (TVD = 0.3888). Only 11.4% of questions favored the matched USA adapter. The degradation is not confined to US data: the USA adapter also worsens fit on Mexico's items (TVD = 0.6377 vs. base 0.5211).

[1] Tier-1 figures derive from the fine-tuning pipeline's cross-evaluation notebook; the committed reproduction of these runs is scheduled (see Section 6).

The mechanism—for example, a collision with the base model's strong American pretraining prior, versus overfitting to the synthetic scenario format—is not identified by this design.

Plain-language reading. "Pretraining prior" refers to the implicit population the model assumes when asked an English-language survey question without country context—a population that looks American. Adding synthetic US signal on top of that prior moves the policy away from the empirical US WVS distribution (TVD 0.396 $\rightarrow$ 0.526); the USA adapter also moves away from Mexico's distribution (0.521 $\rightarrow$ 0.638), so the degradation is not specific to US items. The asymmetry with Mexico is consistent with the prior-collision account, but competing explanations remain viable; the mechanism should be treated as unresolved.

Key Finding 3: Structural Under-Dispersion. Across all models, option log-likelihood scoring suffers from severe **under-dispersion** (dispersion bias ranging from -1.15 to -1.63). Softmax normalization concentrates probability mass on modal choices, underestimating real-world human population variance. Any claim about *absolute shares* ("42% of Mexicans answer X") is unsupported by the current pipeline; claims about *direction and rank* are.

4.3 Construct-validity results

The measurement framework of Section 3.2 produces three families of results.

Admissibility of WVS facets as GPS representations. Table 3 is the headline: only 5 of 16 candidate facets are admissible representations of the six GPS dimensions at the literature-anchored threshold. The failures are *informative*: risktaking's mapped economic-values items do not behave like GPS risk preferences at any resolution (empty M^* even at $\lambda = 0.5$), and the age-shape diagnostic catches financial satisfaction (Q50) as a U-shaped rather than hump-shaped patience proxy—admissible at country level, structurally wrong at individual level.

Instrument equivalence: Q57 vs. the survivor composite. The single-item trust measure and the multi-item composite agree on *average* trust across countries but disagree on *who* has it. At country level, Q57 correlates 0.28 with GPS trust versus 0.41 for the composite. Across 42 countries, the cross-country correlation of the WVS-vs-GPS *gender gaps* is 0.005 for Q57 (no structural alignment) versus 0.568 for the composite (country-cluster bootstrap, $P(\text{composite} > \text{Q57}) = 0.999$). At cell level (sex$\times$age cells within countries), Q57's pooled within-country correlation with GPS trust is $+0.087$ (CI crosses zero) versus $+0.334$ for the composite (CI $[0.234, 0.415]$; robust to GPS weights). The economic-model implication: any model that interacts trust with demographic composition will obtain GPS-inconsistent trust effects when fed Q57—the canonical item used across the empirical trust literature.

Demographic-gradient structure (Stream B). The WVS facets reproduce GPS's positive education/cognitive gradient for the trust facets, patience-financial, and negrecip-justifiability (pooled z between $+4.8$ and $+11.1$), but largely fail GPS's gender gradient—most sharply for trust-general and out-group, where the WVS female coefficient is significantly *negative* where GPS trust is positive. This is direct evidence that GPS trust and WVS generalized trust are related but not interchangeable measures, and it pre-registers which channels the adapter evaluation must reproduce.

5. Discussion

5.1 What the framework establishes

The two contributions answer different questions, and both answers are needed. The pipeline establishes that a measured cultural state can be encoded into a policy such that the resulting fixed response distribution aligns with an external human criterion for populations under-represented in pretraining—an *estimation* result. The measurement framework establishes which survey measures are admissible representations of the incentivized constructs—a *validity* result, with the proviso, following Borsboom et al. [5], that passing a network's restrictions establishes consistency with the network, not causation between attribute and score. The `cvprofiles` instrument makes the validity result reproducible: thresholds anchored in published evidence, slacks reported in full, empty sets reported as findings, and run identities derived from frozen inputs.

5.2 Reliable and unreliable applications

Reliable. (i) **Distributional fidelity shifts**: option-choice probabilities move toward empirical population distributions for populations under-represented in pretraining, measured without country prompts—the adapter's *fixed* distribution is closer to Mexico's observed responses. (ii) **Relative/ordinal comparative statics**: tracking the *direction* of change in a preference dimension across populations, instruments, or experimental arms, not absolute shares. (iii) **Pre-field instrument stress-testing**: probing draft questionnaire items against a target population benchmark before expensive fieldwork; items where model distributions diverge sharply from the benchmark are flagged for translation or scale artifacts. (iv) **Measurement-fragility quantification**: the construct-identified range $[L,U]$ tells the analyst how much of a downstream conclusion is attributable to measurement choice—the H5 example bounds the trust–development correlation between 0.371 and 0.624 across admissible measures.

Unreliable. (i) **Absolute headcount forecasting and polling**: structural under-dispersion makes percentage-share predictions unsupported. (ii) **Fine-tuning over already-dominant priors**: the measured degradation of the USA adapter suggests the right reaction may be to leave the base model alone or use an anchoring regularizer—a hypothesis for the next iteration, not an established conclusion. (iii) **Free-form conversational persona chat**: option-likelihood validation does not certify open-ended dialogue. (iv) **Granular trait elasticities on thin cells**: per-dimension per-question estimates on small cells carry wide confidence intervals and should not be quoted as calibrated structural parameters.

5.3 Frontier applications

The framework opens applications beyond the current pipeline's verified surface; we list them to scope the program.

1. **Power-test analysis without priors.** The construct-identified range $[L,U]$ is a set of plausible effect sizes for the target parameter across admissible measures. A researcher planning a field experiment can compute power curves across the whole range rather than committing to a single prior effect—an honest way to size studies when the construct is latent and measurement choice is contested.
2. **Pre-piloting survey questions for new instruments.** The pipeline can probe draft items through validated adapters and benchmark populations before fielding: items whose model distributions diverge sharply from the target benchmark, or whose implied facet scores violate the network's restrictions, are flagged pre-field. This converts survey design from an expert-judgment task into a falsifiable design loop.
3. **Inducing a preference construct in an arbitrary LLM.** Because the cultural state enters only through the training labels, the pipeline can induce *any* GPS parameterizable disposition in *any* open-weight base model: given a target state z^*, generate pairs under z^*, fine-tune, and validate with the same two-tier protocol. This generalizes from "culture" to arbitrary preference engineering—e.g., inducing a specified

risk attitude in a model that otherwise has none.

4. **Simulating human-subject responses at country or individual level.** The adapters are population simulators: at the *moment* level (country means and option distributions, as in Tier-2) and, via persona-conditioned inference, at the *distribution* level (individual-level response heterogeneity conditional on demographics). The validity ladder of Section 3.2 (country, demographic gradient, cell) transfers to both objects, giving the simulation a defined validity certificate rather than an anecdotal one.

5. **Activation-steered constructs (derivative work).** Steering vectors [30, 31]—directions in activation space that modulate a behavior at inference time—provide a *no-training* route to construct induction: a steering vector for a preference dimension yields a policy whose option distributions are a new measurement function $m_{\text{steer}}(X)$, a candidate in the same menu as m_{base} and m_{DPO}. Because steering intensity is a continuous knob, the menu becomes a continuum and admissibility becomes a function of steering strength; where along the intensity axis a steered measure crosses into the admissible set is a novel, framework-answerable question. This line is planned as a derivative project aimed at a methods/CS venue, not part of the current empirical claims.

6. **A validation feedback loop and a community construct registry.** If the validation protocol works, the network's restrictions become training objectives: fine-tuning can be gated on whether the implied measures satisfy the moment inequalities, yielding *theoretically-aligned* models—policies trained to satisfy a stated nomological network. A public registry where research groups contribute construct definitions (score matrices plus nomological networks, subject to the discipline of anchored thresholds and frozen runs) would let measurement validity accumulate as a community resource rather than per-paper judgment.

5.4 Positioning: measurement and developing economics

Economics has always studied latent constructs. Trust, patience, risk attitudes, and social preferences are not directly observable; they are recovered through instruments—surveys, incentivized tasks, experimental games—each of which is a costly, imperfect representation. The field's standard response is to *justify a single proxy* and proceed. That response is increasingly untenable: the canonical trust item and a multi-item composite disagree on *who* trusts, and the choice between them moves the trust–development correlation by a quarter of a correlation point. The argument of this paper is that the binding constraint on empirical work with latent constructs is no longer the cost of building measures but the discipline of *choosing among them*. AI collapses the cost of candidate measures; partial identification over a nomological network supplies the discipline; the two together give developing economics what it has lacked since Cronbach and Meehl: a reproducible procedure for saying which measures are admissible and what conclusions survive all admissible ones. The implications run from hypothesis generation (survey new items before fielding) to cross-country policy design (which representation drives the policy conclusion—and what set of conclusions is robust to representation choice).

5.5 Limitations

The Tier-2 evaluation compares adapter outputs to human survey responses—an external, non-AI criterion—which answers the correlated-errors objection to validating AI measures against AI measures [14]. Three caveats remain. First, the adapters and their synthetic training labels share a common generative ancestry, so Tier-1 self-recovery is in-sample evidence only. Second, the mechanism behind the USA adapter's degradation is unresolved. Third, the country panel is two adapters and two evaluation countries; the measurement layer runs on 42 countries, but the adapter layer does not. The ecological caveat applies throughout: country-level correlations are development-confounded, which is why the demographic-gradient and cell-level resolutions are load-bearing rather than decorative.

6. Next Steps

The agenda below is the minimal credible next iteration. **Stream A (measurement):** fold the P1, demographic-gradient, and cell-mean layers into single multi-restriction `cvprofiles` networks per dimension (adding discriminant and known-groups restrictions), and adopt a downstream target β (e.g., log-GDP correlation) so that every dimension reports a construct-identified range. **Stream B (persona gradients):** run the frozen pre-registered persona-conditioned evaluation—the designed adjudication of whether the adapters encode demographic structure—with the survivor composite as the WVS-side reference. **Stream C (falsification):** add a third country (Argentina) whose GPS profile differs from Mexico's to test the prior-collision account, growing the adapter panel at the current training-data scale—intentionally capped to avoid new dataset generation; and reframe the AmericasBarometer surface as an instrument-redundancy check (its local coverage is USA/MEX only). Adapter *mixtures* (convex combinations of adapter weights or output distributions) are themselves candidate measures: the validity layer can test whether intermediate GPS states are admissible, treating interpolation as a measurement-construction device. **Stream D (calibration):** address under-dispersion via per-question calibration and temperature sensitivity. **Stream E (provenance):** reproduce and commit the Tier-1 cross-evaluation outputs.

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